

ME8930: Data-driven Learning and Control of Dynamic Systems:

Assignment 2

Course GitHub: https://github.com/DyCo-AI/data_driven_learning_course

Problem 1.

Consider the one dimensional dynamical system of the form

$$\dot{x} = -x + x^3.$$

1. Perform the Koopman lifting of system using the basis functions $\{x^{2n+1}\}$ for $n = 0, 1, 2, \dots, N$.
2. Write a MATLAB code to compute the finite dimensional approximation of the Koopman generator for any finite N .
3. Verify that the eigenfunction of the Koopman generator for this system is given by $\frac{x}{\sqrt{1-x^2}}$.
4. Compute the eigenfunction of the Koopman generator for changing N and compare it with the analytical expression of the eigenfunction for changing N .

Problem 2.

Consider the dynamics of a damped simple pendulum given by

$$\ddot{\theta} + b\dot{\theta} + \frac{g}{L} \sin \theta = 0.$$

Assume the pendulum has unit mass and unit length ($L = 1$) with damping coefficient $b = 2$.

1. Write the above system in a state space form and find the eigenvalues of this system around the stable equilibrium point at $(0, 0)$.
2. Load the training dataset from `training\pendulum_training.csv`. This dataset consists of 126 trajectories collected with a time span of 1 s and a sampling rate of 0.1 s for each trajectory. Visualize a few sample trajectories by plotting a phase portrait (θ vs $\dot{\theta}$).
3. Obtain the finite-dimensional approximation for the Koopman operator using Dynamic Mode Decomposition (DMD). Plot the eigenvalues of the operator in a unit circle. For this, you need to complete the code in `DMD\get_DMD.m` function and call it in the main file (`simple_pendulum.m`) for further analysis.
4. Lift the data using a monomial basis $\psi(\mathbf{x})$ and obtain the finite-dimensional approximation for the Koopman operator using Extended Dynamic Mode Decomposition (EDMD). Note that you have to append the original states $\mathbf{x}$ to obtain $\Psi = [\mathbf{x}, \psi(\mathbf{x})]$ to obtain

$$\mathbf{C} = \begin{bmatrix} \mathbb{I}_{n \times n} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}.$$

Plot the eigenvalues of the operator in a unit circle. For this, you need to complete the `EDMD\get_EDMD.m` function and call it in the main file (`simple_pendulum.m`) for further analysis.

5. Load the validation dataset from `prediction/pendulum_validation.csv`. This dataset consists of 9 trajectories collected with a time span of 10 s and a sampling rate of 0.1 s for each trajectory. Visualize it by plotting a phase portrait.
6. Compare the prediction accuracy of the Koopman operator obtained using DMD and EDMD on the validation dataset using root mean square error (rmse) and % error.
 - DMD: use $\hat{\mathbf{x}}_{t+1} = \mathbf{K}\mathbf{x}_t$. Here $\hat{\mathbf{x}}$ is the one time-step prediction.
 - EDMD: use $\hat{\mathbf{z}}_{t+1} = \mathbf{K}\mathbf{z}_t$. Here $\hat{\mathbf{z}}_t$ is the one time-step prediction in the lifted state. To obtain the original state prediction, use $\hat{\mathbf{x}}_t = \mathbf{C}\hat{\mathbf{z}}_t$.

For this, you need to complete the code in `prediction\eval_prediction.m` function to output `X_pred`, which contains trajectories obtained by stacking $\hat{\mathbf{x}}_t$ for $n = 20$ time-steps. Then, use the `prediction\rmse.m` function to evaluate rmse and % error.

Problem 3.

Consider the dynamics of a duffing oscillator given by

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_1 - \delta x_2 - x_1^3\end{aligned}$$

where $\delta = 0.5$.

1. Load the training dataset from `training\duffing_training.csv`. This dataset consists of 100 trajectories obtained from initial conditions sampled using a uniform grid with the domain $[-2, 2]$. Each trajectory is collected with a time span of 1 s and a sampling rate of 0.1 s for each trajectory. Visualize a few sample trajectories by plotting a phase portrait (x_1 vs x_2).
2. Lift the data using a monomial basis $\psi(\mathbf{x})$ and obtain the finite-dimensional approximation for the Koopman operator using Extended Dynamic Mode Decomposition (EDMD). Plot the eigenvalues of the operator in a unit circle.
3. Load the validation dataset from `prediction\duffing_validation.csv`. This dataset consists of 9 trajectories collected with a time span of 10 s and a sampling rate of 0.1 s for each trajectory. Note that all trajectories lie within the basin of attraction of the equilibrium point $(1, 0)$. Visualize it by plotting a phase portrait.
4. Find the prediction accuracy of the Koopman operator obtained using EDMD on the validation dataset using root mean square error (rmse) and % error.
5. Now, load the training dataset from `training\duffing_multi_model_training.csv`. This dataset consists of 100 trajectories that lie within the domain of attraction of the equilibrium point $(1, 0)$. Each trajectory is collected with a time span of 1 s and a sampling rate of 0.1 s for each trajectory. Visualize a few sample trajectories by plotting a phase portrait (x_1 vs x_2).
6. Lift the data using a monomial basis $\psi(\mathbf{x})$ and obtain the finite-dimensional approximation for the Koopman operator using Extended Dynamic Mode Decomposition (EDMD). Plot the eigenvalues of the operator in a unit circle.

7. Compare the prediction accuracy of the two Koopman operators obtained on the same validation dataset using root mean square error (rmse) and % error.

Problem 4.

Consider a dynamical system of the form

$$\dot{x} = \begin{bmatrix} \frac{(7.5x_2^2+5.0)(x_1^3+x_1+\sin(x_2))+(-x_1+x_2^3+2x_2)\cos(x_2)}{9x_1^2x_2^2+6x_1^2+3x_2^2+\cos(x_2)+2} \\ \frac{2.5x_1^3+2.5x_1-(3x_1^2+1)(-x_1+x_2^3+2x_2)+2.5\sin(x_2)}{9x_1^2x_2^2+6x_1^2+3x_2^2+\cos(x_2)+2} \end{bmatrix}. \quad (1)$$

1. Verify that the principal eigenfunction associated with eigenvalues $\lambda_1 = -1$ and $\lambda_2 = 2.5$ are given by

$$\begin{aligned} \phi_{\lambda_1} &= x_1 - 2x_2 - x_2^3, \quad \lambda_1 = -1 \text{ and} \\ \phi_{\lambda_2} &= x_1 + \sin(x_2) + x_1^3, \quad \lambda_2 = 2.5. \end{aligned}$$

2. You may use the `compute_eigfn.m` as the main file and the convex optimization code provided in `eigenfunctions/convex_opt.m` to approximate the principle eigenfunctions of this system and compare with part 1.
3. Compute the zero-level curves of the eigenfunctions to identify the stable and the unstable manifolds of the equilibrium point at the origin.